

Part B:

Date: _____

Q1) PAC Chart:

* Given Data

- Season, Room Type
- Standard = 5000 PKR/night
- Deluxe = 8000 PKR/night
- Suite = 12,000 PKR/night

* Required Results

- Final Price for each guest
- Total Revenue

* Processing Required

$$\text{Total Price} = (\text{rate} \times \text{nights}) - \text{discount}$$

* Solution Alternatives

- Use Arrays to store each guests Price

→ IPO Chart

Input

- N guests
- Season and Room Type

Process

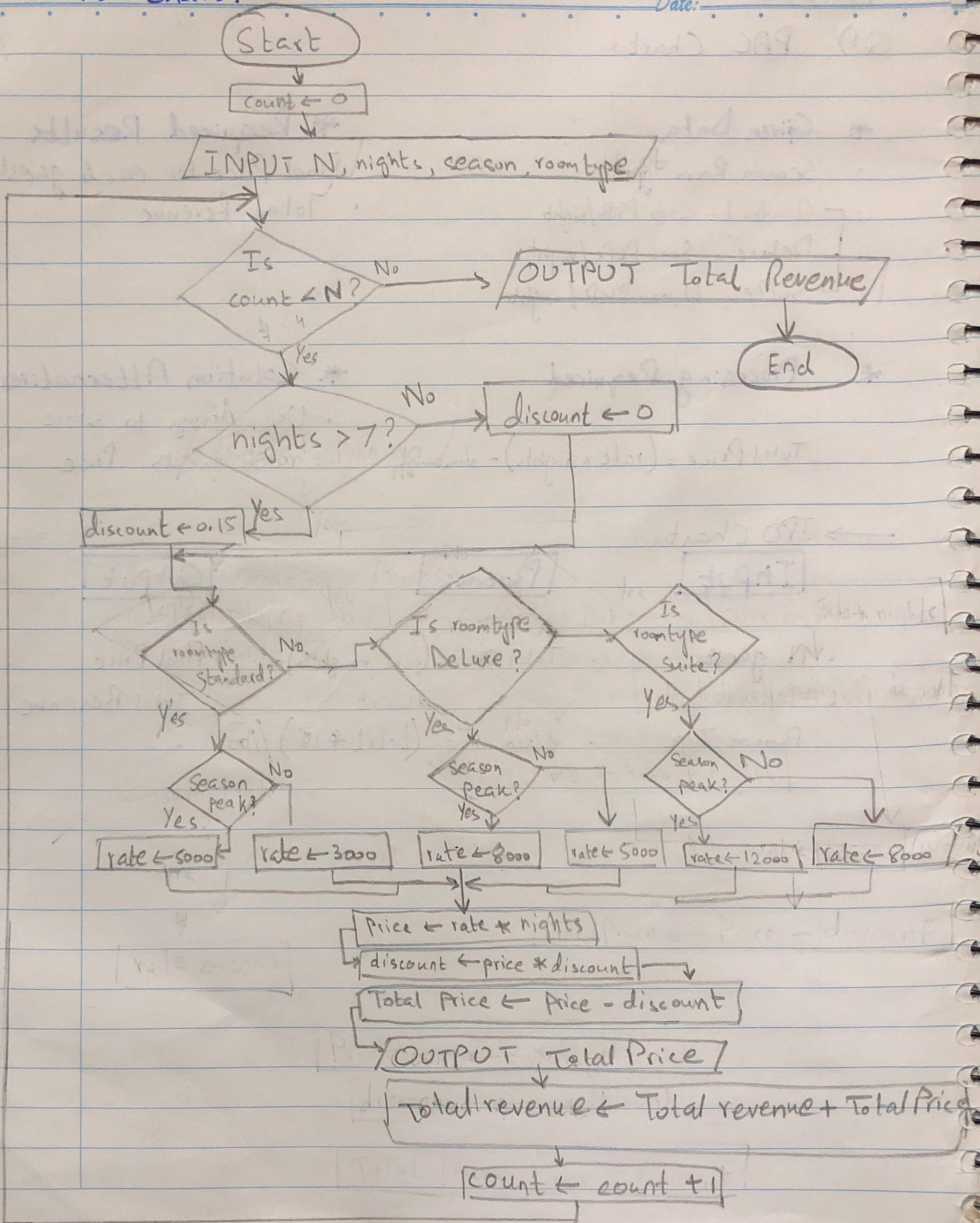
- $\text{Total Price} = (\text{rate} \times \text{nights}) - \text{discount}$
- $\text{discount} = (\text{total} * 15) / 100$

Output

- Final Price
- Total Revenue

→ Flow chart.

Date: 8/1/2019



→ Pseudocode.

Date: _____

```

DECLARE count, rate, nights, N : INTEGER
DECLARE discount, price, Total_Price, Total_Revenue : REAL
count ← 0, discount ← 0, rate ← 0, price ← 0, Total_Price
INPUT N, nights,
INPUT season, roomtype
FOR count 0 TO N-1
    IF night > 7 THEN
        discount ← 0.15
    ENDIF
    IF roomtype = "standard" THEN
        IF season = "Peak" THEN
            rate ← 5000
        ELSE
            rate ← 3000
        ENDIF
    ELSE
        IF roomtype = "Deluxe" THEN
            IF season = "Peak" THEN
                rate ← 8000
            ELSE
                rate ← 5000
            ENDIF
        ELSE
            IF roomtype = "Suite" THEN
                IF season = "Peak"
                    rate ← 12,000
                ELSE
                    rate ← 8000
                ENDIF
            ENDIF
        ENDIF
    ENDIF

```

Date: _____

LENDIF

 $\text{Price} \leftarrow \text{rate} * \text{nights}$ $\text{discount} \leftarrow \text{price} * \text{discount}$ $\text{Total_Price} \leftarrow \text{Price} - \text{discount}$

OUTPUT Total_Price

 $\text{Total_Revenue} \leftarrow \text{Total_Revenue} + \text{Total_Price}$

NEXT count

OUTPUT "The total revenue generated is," Total_Revenue